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Turbine engine centerbody generator

Abstract

A gas turbine engine has: a compressor; a combustor; a turbine; a generator in a centerbody and coupled to the turbine; a case; a plurality of struts mounting the centerbody to the case; and wiring passing through at least one of the struts to the generator. The centerbody has: a first member having a forward rim and at least one recess extending aft from an opening at the forward rim, the wiring passing through the recess; at least a second member blocking the opening; and a threaded fastener securing the strut to the second member.

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Background/Summary

BACKGROUND

(1) The disclosure relates to gas turbine engines. More particularly, the disclosure relates to a centerbody generator.

(2) Gas turbine engines (used in propulsion and power applications and broadly inclusive of turbojets, turboprops, turbofans, turboshafts, industrial gas turbines, and the like) include generators.

(3) An example engine is a multi-spool engine such as a two-spool engine. A two-spool engine will have a high speed/pressure spool and a low speed/pressure spool. In an example wherein each spool has a compressor section and a turbine section, the gaspath passes sequentially through the low speed/pressure compressor (LPC) section, the high speed/pressure compressor (HPC) section, the combustor (e.g., annular or can-type), the high speed/pressure turbine (HPT) section, and the low speed/pressure turbine (LPT) section. In an example two-spool turbofan engine, the fan is upstream/forward of the LPC and is directly powered by the low spool. The HPT drives rotation of the HPC and the LPT drives rotation of the LPC and fan. In alternative turbofan embodiments, there may be no separate LPC with the LPT driving only the fan. In other embodiments, a reduction transmission (e.g., epicyclic) may intervene between the low spool and the fan to drive

the fan with a gear reduction relative to the LPT and thus allow a larger diameter fan than might otherwise be achieved.
(4) In such multi-spool engines, a generator is often run off the high spool with a gear on the high spool driving a shaft passing radially outward to a generator and/or other accessories typically mounted near the bottom of the engine well offset from the engine centerline.
(5) Despite the prevalence of such offset generators, coaxial generators mounted to one of the spools have been proposed.
(6) U.S. Pat. No. 10,605,165, “Aircraft Engine Apparatus” of Abe et al., Mar. 31, 2020 discloses a generator mounted in a nose cone of a turbofan engine.
SUMMARY
(7) One aspect of the disclosure involves a gas turbine engine comprising: a compressor; a combustor; a turbine; a generator in a centerbody and coupled to the turbine; a case; a plurality of struts mounting the centerbody to the case; and wiring passing through at least one of the struts to the generator. The centerbody comprises: a first member having a forward rim and at least one recess extending aft from an opening at the forward rim, the wiring passing through the recess; at least a second member blocking the opening; and a threaded fastener securing the strut to the second member.
(8) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the centerbody further comprises a third member wherein: the third member has an external thread mated to an internal thread of the first member; and the threaded fastener locks the third member against unthreading rotation.
(9) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the threaded fastener is in threaded engagement with a hole in the second member and passes through unthreaded holes in the strut and third member so as to lock the third member against unthreading rotation.
(10) A further embodiment of any of the foregoing embodiments additionally and/or alternatively, the third member is a nose cone.
(11) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the second member has a pair of projections extending into the opening at opposite circumferential sides thereof.
(12) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include a fan having a fan hub wherein: a drive hub couples a rotor of the generator to the fan hub; and the fan hub couples the drive hub to the turbine.
(13) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the fan is driven by the turbine without reduction.
(14) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the turbine is a low-speed turbine section and the engine includes a high speed turbine section upstream of the low speed turbine section along a gaspath.
(15) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include a second generator coupled to the turbine.
(16) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the turbine comprises a higher pressure turbine and a lower pressure turbine, the lower pressure turbine downstream of the higher pressure turbine along a gaspath; the second generator is coupled to the higher pressure turbine; and the generator is coupled to the lower pressure turbine.
(17) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the second generator is coupled to the turbine via a radially-extending shaft.
(18) A further aspect of the disclosure involves a method for manufacturing the gas turbine engine. The method comprises: inserting the generator into the first member; mating the second member to the first member; threading a third member to the first member to axially retain the second member

to the first member; and installing the threaded fastener to secure the strut to the second member and lock the third member against unthreading rotation.
(19) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the mating causes protrusions on the second member to mate with the first member to secure the second member to the first member against rotation.
(20) A further aspect of the disclosure involves a method for using the gas turbine engine. The method comprises: running the engine so that the turbine drives a fan having a fan hub; and the fan hub driving the generator.
(21) A further aspect of the disclosure involves, a gas turbine engine comprising: a compressor; a combustor; a turbine; a generator in a centerbody and coupled to the turbine; a case; a plurality of struts mounting the centerbody to the case; and wiring passing through at least one of the struts to the generator. The centerbody comprises means for receiving the generator with the wiring attached while limiting folding/bending the wiring.
(22) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the means comprises: a first member having a rim and at least one recess extending from an opening at the rim, the wiring passing through the recess; and at least a second member blocking the opening.
(23) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively, the struts are secured to both the first member and the second member.
(24) A further aspect of the disclosure involves a method for assembling a centerbody, the method comprising: inserting a generator into a first section of the centerbody; mating a ring to a forward end of the first section; threading a nose cone to the first section to axially retain the ring to the first section; and installing a threaded fastener to secure a strut to the ring and lock the nosecone against unthreading rotation.
(25) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include coupling a rotor of the generator to a fan hub for driving the rotor.
(26) In a further embodiment of any of the foregoing embodiments, additionally and/or alternatively: the engine has at least three said struts and the installing is of a respective a threaded fastener to secure each said strut to the ring and lock the nosecone against unthreading rotation; and for at least one of the struts, radially inserting the strut through a case so as to receive wiring from the generator.
(27) The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic half sectional view of a turbine engine with centerbody/nose cone generator.
(2) FIG. **2** is a view of an auxiliary generator section of the engine.
(3) FIG. **3** is a front view of the auxiliary generator section.
(4) FIG. **4** is a transverse sectional view of the auxiliary generator section taken along line **4-4** of FIG. **3**.
(5) FIG. **4**A is an enlarged view of strut mounting.
(6) FIG. **4**B is a further enlarged view of strut mounting.
(7) FIG. **5** is a transverse sectional view of a strut taken along line **5-5** of FIG. **4**.
(8) FIG. **6** is a view of the auxiliary generator section along the FIG. **4** cut plane during a first intermediate stage of assembly.

(9) FIG. **7** is an inward radial view of the centerbody of the auxiliary generator section of FIG. **6**.
(10) FIG. **8** is a view of the auxiliary generator section along the FIG. **4** cut plane during a second intermediate stage of assembly.
(11) FIG. **9** is an inward radial view of the centerbody of the auxiliary generator section of FIG. **8**.
(12) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(13) FIG. **1** shows a gas turbine engine **20** including a main engine section **22** and an accessory generator section **24**. The example main engine section **22** is shown as a two-spool turbofan having, sequentially along a gaspath **900**, from upstream to downstream, a fan section **30** having a circumferential array of fan blades **32**, a low pressure compressor (LPC) section **40**, a high pressure compressor (HPC) section **42**, a combustor section **44**, a high pressure turbine (HPT) section **46**, and a low pressure turbine (LPT) section **48**. Each of the LPC, HPC, HPT, and LPT includes one or more stages of blades interspersed with one or more stages of vanes. The blades of the respective sections are coupled to a respective high speed shaft **52** or low speed shaft **50**. The shafts are supported to each other and/or static case structure **60** via one or more bearing systems (not shown) to allow each of the low spool and high spool to rotate about an engine centerline or central longitudinal axis **500**.
(14) In the example, the fan **30** is co-spooled on the low spool so as to be directly driven by the low shaft **50**. In alternative embodiments, a transmission (not shown, e.g., epicyclic) may intervene so as to drive the fan with a gear reduction. Yet alternative embodiments involve other configurations of one or more spools. The example fan blades are surrounded by a fan case **62** supported relative to the core case **60** via a circumferential array of struts **64** across a bypass flowpath **902** diverging from the core flowpath **900** at a splitter **66**.
(15) As discussed further below, it may be desirable to incorporate a generator **100** into a centerbody structure **102**. The generator rotor **104** may be coaxial with and driven by one of the engine spools to rotate relative to a generator stator **106**. Depending upon implementation, this generator **100** may replace or augment an existing offset (e.g., driven via a tower shaft and gear system) generator **80** (FIG. **1**). In one group of examples, there may be a requirement for power generation beyond what is offered by a baseline HPT-driven offset generator **80** (driven via a tower shaft **81** from a gear **82** on the high speed shaft). This may be in a situation wherein packaging does not allow any substantial increase in the size of the baseline offset generator. Or the HPT may be insufficient to power the increased load.
(16) In one group of such examples discussed below, the normal nose cone/spinner of a turbofan engine is removed and replaced with a stationary centerbody **102** containing the added generator **100**. The generator rotor **104** is mechanically secured relative to the fan to be driven by the same turbine section driving the fan (e.g., the LPT in the multi-spool example). To hold the centerbody **102** stationary (against rotation), the centerbody may be supported relative to a case structure **120** (e.g., a case ring insert/extension discussed below) via a number of radial struts **122** extending across the now-extended inlet flowpath leg (upstream of the core v. bypass split).
(17) For example, for a wing nacelle turbofan engine, the leading inlet cowl ring structure of the baseline engine nacelle (fan case nacelle) may be removed and a case ring insert/extension **120** (bearing the struts **122** supporting the added centerbody **102**) may be mounted to the forward end of the existing fan case structure **62**. For example, the case ring insert **120** may have, at an aft rim, mounting features **128** complementary to the existing forward rim mounting features **129** of the fan case (which had been used to mount the inlet cowl). In a simple implementation, these may merely be a bolt circle or other arrangement (e.g., with associated mounting ears on each of the case ring insert and fan case for receiving threaded fasteners). A replacement inlet cowl may mount to the case ring insert to slightly extend the fan case/nacelle. In one example, the forward rim of the case ring insert **120** may have mounting features **127** identical to those **129** of the forward rim of the fan case structure to mount the re-used or a new replacement inlet cowl. If the inlet cowl is re-used, an

outer diameter cowl insert would bridge aft therefrom to the remaining existing cowl structure. FIG. **1** shows an overall final cowl outline in broken line.

(18) In the example, the generator rotor **104** is supported via a shaft **130** supported by bearings **132** for rotation about the engine centerline **500** relative to the stator **106**. In the particular example, the generator bearings mount to a generator case structure/housing assembly **134** (although schematically shown as a single piece). The example forward bearing **132** may be supported via a spider structure (not shown) so as to expose the rotor and stator to cooling air. A rear input end section **140** of the shaft **130** may bear features such as splines for engaging complementary features of a forward end of a drive hub **142** whose aft end is mounted to the baseline fan hub **143** (e.g., via the same or similar bolt circle of fasteners (not shown or other features) that were used to mount the baseline nose cone to the fan hub). The drive hub **142** is largely within an aft section of a main housing piece **150** of the centerbody extending to an aft rim **152** spaced slightly forward of the fan hub to allow relative differential rotation as the fan rotates. As is discussed further below, the main piece **150** extends aft from a forward rim **151** (FIG. **7**). The centerbody housing also includes a nose cone piece **160** and an intermediate piece **170**. The main piece **150** (FIG. **4**) has an outer diameter (OD) surface **153** and an inner diameter (ID) surface **154**. At an intermediate location along its length, the main piece **150** includes a radially inwardly extending mounting flange or mounting ears **155** (FIG. **4**A) for mounting the generator (e.g., via fasteners **157** through an external flange or mounting ears **156** of the generator housing **134**).

(19) The diametric size of the generator stator relative to that of the centerbody is an important consideration. In general, aerodynamic efficiency, packaging efficiency, and weight favor a small diameter centerbody. However, generator capacity may favor a radially large generator. Minimizing radial clearance between the generator stator and the centerbody inner diameter (ID) surface is thus desirable. Control/monitoring wiring and/or power wiring influence packaging considerations. Inserting the generator through a rim of a centerbody housing section during installation presents issues of potentially bending the wiring in order to then feed the wiring outward through apertures in the centerbody (which apertures will subsequently be surrounded by struts) to pass the wiring radially outward. Thus, the available bend radius of the wiring imposes a radial clearance requirement that means the stator outer diameter would be otherwise excessively smaller than the housing inner diameter.

(20) By making one or more apertures **180** in that centerbody main housing piece **150** open to the forward rim **151**, the generator may be installed without the need to bend the wiring. Thus, with wiring protruding radially from the stator, the stator may pass into the main housing piece **150** with little generator-to-centerbody radial clearance and the wiring passing through the opening of the aperture(s) in the forward rim. This may allow very close radial accommodation between the generator outer diameter (OD) surface and the centerbody section inner diameter (ID) surface. Those open apertures may then be closed off by one or more other centerbody housing pieces. FIG. **7** shows such an aperture **180** having an open end **182** at the forward rim **151**. The aperture **180** is formed at a strut mounting feature **184** (shown as a protruding mounting boss with footprint generally similar to the strut footprint). FIG. **7** also shows the wiring harness **190** about to pass into the aperture **180** opening **182**.

(21) After the generator has been installed, it may be secured in place. Depending upon implementation, this may involve inserting fasteners from the rear or the front. In the illustrated example, screws **157** (FIG. **6**) are inserted through the front to pass through unthreaded holes in the flange or ears **156** and to engage threaded holes in the flange **155**. After the generator is secured in place, the intermediate piece **170** may be installed. The example intermediate piece **170** has a forward rim **171**, an aft rim **172**, an outer diameter (OD) surface **173**, and an inner diameter (ID) surface **174**. To close off the aperture **180**, the intermediate piece **170** includes a protruding boss feature **184-1** (FIG. **7**) complementary to a boss feature **184-2** of the main piece **150** for forming the strut mounting feature **184**. The boss feature **184-1** has a pair of rearwardly protruding

projections **185** received in the aperture **180** to angularly register the pieces **170** and **150** with each other. The example main piece **150** has, at its forward rim **151**, a radially inwardly recessed shoulder **158** dimensioned to be received within an aft portion of the intermediate piece **170** in close or contacting radial proximity.

(22) The example nose cone piece **160** has a forward end **161**, an aft rim **162**, an OD surface **163**, and an ID surface **164** (FIG. **6**). The example nose cone piece has a central venting opening **165** at a rim **166**. An aft portion of the nose piece near the aft rim **162** is stepped radially inward with an externally threaded rear section **200** and a section **202** forward thereof having a circumferential array of holes **204**. After assembly of the intermediate piece **170** to the main piece **150**, the nose cone piece **160** may be inserted and rotated to thread into place. The threading engagement is with an internally threaded section **210** of the main piece **150** near the rim **182**.

(23) As is discussed further below, retention of the nose cone piece **160** against unthreading is provided by an example forward one **220-1** (FIG. **4**A) of two fasteners **220-1**, **220-2** used to mount each strut to the centerbody housing assembly. FIG. **8** shows, after assembling the centerbody and placing the case **120** in position, each of the struts **122** may be radially inwardly inserted through an aperture **224** in the case **120** (e.g., surrounded by a boss **226**). This may be preceded by feeding the wiring through the associated aperture in the case. The radial clearance between case and centerbody means only slight bending of the wiring is needed. For example bending radius may be not less than an example ten multiples of the wire/bundle diameter or an example not less than six times or not less than four times) Example wire/bundle diameters are 3.0 millimeters to 30.0 millimeters or 8.0 millimeters to 25.0 millimeters. The example strut, at its outer diameter end **240** has an outwardly protruding flange **228** which contacts the apertured boss **226** on the case **120**. During strut installation, the inward radial end (ID end) **242** of the strut has an opening passing the wiring **190** (ultimately out an opening at the OD end **240**. At the ID end **242**, the strut has respective fore and aft apertured interior mounting flanges **243** for passing the screws **220-1** and **220-2**, respectively. To receive these screws, the boss section **184-2** has a threaded aperture **244-2** (FIG. **7**) so that the screw **220-2** passes freely through the strut aft mounting aperture to thread into that threaded. boss mounting aperture. In this example, the intermediate piece boss section **184-1** also has a threaded aperture **244-1**. However, the screw **220-1** is longer than the screw **220-2** having a narrowed unthreaded tip portion capable of protruding well past the ID surface **174** of the intermediate piece and into a locally adjacent one of the nose cone holes **204**. The receipt in such hole **204** secures the nose cone against rotation (e.g., unthreading rotation). The fasteners **220-1** and **220-2** may be tightened via a socket wrench using an extension extending radially through the strut from the OD end thereof.

(24) In the illustrated example, all three of the struts **122** pass associated wiring. However, this need not be the case. As few as one strut may pass such wiring. The other struts and their mounting features may be otherwise the same as the one passing wiring. Optionally, one or more of the struts may additionally or alternatively pass some other utility such as a cooling and/or lubricating liquid. Optionally, if a strut is not passing wiring, fluid, etc., its interior may be closed off such as by a cover plate (not shown).

(25) In use, the generator **100** may replace, supplement, or complement the generator **80** (when both are present). For example, in certain aircraft with added loads over a baseline aircraft, the generator **100** may exclusively or principally power such added loads. This may be relevant, for example, to aircraft with updated or added avionics (e.g., including sensors, communications hardware, and the like) or other added loads. In replacement situations, there may not be enough space to provide a large enough generator **80** to power all loads but there may be enough space for a generator **100** large enough to do so. This replacement is more likely in a new engine build situation than in a retrofit. A given basic engine may be used in diverse applications with diverse electrical power requirements. Thus, the generator **100** may be selected to meet the power requirements for any particular vehicle that uses a given engine. The form of generator **100** may be

selected based upon the type of load. On example for many loads is a synchronous AC generator. There may be additional power regulation, conversion (step-up or step-down or AC-DC or DC-AC), and/or storage (e.g., batteries/or capacitors) equipment (not shown).

(26) Component materials and manufacture techniques and assembly techniques may be otherwise conventional. Additionally, example materials for the housing pieces **150**, **160**, **170** are alloys such as steel or aluminum alloys or titanium alloys and example manufacture techniques are machining and/or additive manufacture. Example materials and manufacture techniques for the struts **122** are likewise. Example materials and manufacture techniques for the case **120** are likewise, but weldments start to become viable. And, theoretically, composites are also possibilities for all of the foregoing.

(27) The use of "first", "second", and the like in the following claims is for differentiation within the claim only and does not necessarily indicate relative or absolute importance or temporal order. Similarly, the identification in a claim of one element as "first" (or the like) does not preclude such "first" element from identifying an element that is referred to as "second" (or the like) in another claim or in the description.

(28) One or more embodiments have been described. Nevertheless, it will be understood that various modifications may be made. For example, when applied to an existing baseline engine or aircraft configuration, details of such baseline may influence details of particular implementations. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A gas turbine engine comprising: a compressor; a combustor; a turbine; a generator in a centerbody and coupled to the turbine; a case; a plurality of struts mounting the centerbody to the case; and wiring passing through at least one of the struts to the generator, wherein: the centerbody comprises: a first member having a forward rim and at least one recess extending aft from an opening at the forward rim, the wiring passing through the at least one recess; at least a second member in contact with the first member such that the opening is blocked; a third member; and a threaded fastener securing the strut to the second member, wherein: the third member has an external thread mated to an internal thread of the first member; the threaded fastener locks the third member against unthreading rotation; and the threaded fastener is in threaded engagement with a hole in the second member and passes through unthreaded holes in the at least one strut and the third member so as to lock the third member against unthreading rotation.
2. The gas turbine engine of claim 1 wherein: the wiring passes through all struts of the plurality of struts.
3. The gas turbine engine of claim 2 wherein: the plurality of struts is three struts.
4. The gas turbine engine of claim 1 wherein: the third member is a nose cone.
5. The gas turbine engine of claim 1 wherein: the second member has a pair of projections extending into the opening at opposite circumferential sides thereof.
6. The gas turbine engine of claim 1 further comprising a fan having a fan hub wherein: a drive hub couples a rotor of the generator to the fan hub; and the fan hub couples the drive hub to the turbine.
7. The gas turbine engine of claim 6 wherein: the fan is driven by the turbine without reduction.
8. The gas turbine engine of claim 7 wherein: the turbine is a low-speed turbine section and the gas turbine engine includes a high-speed turbine section upstream of the low-speed turbine section along a gas path.
9. The gas turbine engine of claim 1 further comprising: a second generator coupled to the turbine.
10. The gas turbine engine of claim 9 wherein: the turbine comprises a higher pressure turbine and a lower pressure turbine, the lower pressure turbine downstream of the higher pressure turbine along a gas-path; the second generator is coupled to the higher pressure turbine; and the generator is coupled to the lower pressure turbine.

11. The gas turbine engine of claim 9 wherein: the second generator is coupled to the turbine via a radially-extending shaft.
12. A method for manufacturing the gas turbine engine of claim 1, the method comprising: inserting the generator into the first member; mating the second member to the first member; threading the third member to the first member to axially retain the second member to the first member; and installing the threaded fastener to secure the strut to the second member and lock the third member against unthreading rotation.
13. The method of claim 12 wherein: the mating causes protrusions on the second member to mate with the first member to secure the second member to the first member against rotation.
14. A method for using the gas turbine engine of claim 1, the method comprising: running the gas turbine engine so that the turbine drives a fan having a fan hub; and the fan hub driving the generator.
15. The method of claim 14, wherein the gas turbine engine further comprises: a second generator coupled to the turbine.
16. The method of claim 15 wherein: the turbine comprises a higher pressure turbine and a lower pressure turbine, the lower pressure turbine downstream of the higher pressure turbine along a gas-path; the second generator is coupled to the higher pressure turbine; and the generator is coupled to the lower pressure turbine.
17. The method of claim 16 wherein: the second generator is coupled to the turbine via a radially-extending shaft.